“Travel Assistant chatbot”

Final Project (CSE2009) Report Submitted to VIT-AP University in Partial Fulfilment for the Award of the Degree of Bachelor of technology
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By

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Abstract:

This project presents the design and implementation of an intelligent Travel Assistant Chatbot for personalized destination recommendations in India. The system confronts the challenge of ambiguous and imprecise user preferences by employing a **fuzzy logic** inference engine built with Python's skfuzzy library. The chatbot, which uses a **Gradio** framework for its stateful, conversational interface, guides users to provide their preferences for **budget**, **climate**, and **interest level**. These inputs are processed by a fuzzy matching system that ranks 50 curated destinations, each acting as an implicit fuzzy rule. The system's "defuzzification" method is a winner-take-all ranking based on an aggregated membership score. Key features include real-time data integration via the **OpenWeatherMap API** for live weather and forecasts, and a session-based adaptive learning mechanism, simulate_backpropagation, which dynamically adjusts recommendation scores based on negative user feedback. The result is a robust prototype that effectively models linguistic ambiguity, demonstrating a practical application of fuzzy logic as a multi-criteria decision-support tool for the travel domain.

Index:

S.no	Title	Page. No
1	Introduction	4
2	Background	4
3	Problem Definition	5
4	Objectives	5
5	Methodology	6-7
6	Results and Discussion	9-10
7	Conclusion and Future scope	11
8	References	11
9	Codes in appendix	12-28

List of Tables:

S.no	Name of figure	Page. No
2	Software Libraries and their Functions	8

I. Introduction:

The task of planning travel is often a complex, multi-criteria decision-making problem. A traveler must balance numerous factors, such as budget constraints, preferred weather conditions, and personal interests (e.g., adventure, heritage, relaxation). Compounding this, these preferences are often not precise; a user may have a budget "around ₹50,000" or desire a climate that is "cool" but not "freezing."

This project addresses this gap by creating an intelligent chatbot that leverages **fuzzy logic**. Fuzzy logic is a computational approach based on "degrees of truth" rather than the usual "true/false" Boolean logic. It provides an excellent framework for modeling linguistic ambiguity.

By building the system as a chatbot using the Gradio framework, we create a natural, conversational, and stateful user experience. The user is guided through a series of questions, and their inputs are processed by a fuzzy inference engine to determine the single best recommendation. The system is further enhanced with real-time API data and an adaptive feedback loop, creating a tool that is both intelligent and practical.

II. Background:

Fuzzy Logic, introduced by Lotfi Zadeh in 1965, is a form of many-valued logic that deals with reasoning that is approximate rather than exact. It is particularly suited for modeling human language and decision-making. Key concepts used in this project include:

- **Linguistic Variables:** Variables whose values are words or sentences in a natural language, not numbers. In this project, budget, climate, and interest are treated as linguistic variables.
- **Fuzzy Sets:** Sets whose elements have degrees of membership. For example, a budget of ₹45,000 might have a 0.8 degree of membership in the "Medium Budget" set and a 0.3 degree of membership in the "High Budget" set.
- **Membership Functions:** A curve that defines how each point in the input space (the "universe of discourse") is mapped to a membership value (or "degree of truth") between 0 and 1. This project dynamically generates **Triangular Membership Functions (trimf)** for each destination's attributes.
- **Fuzzy Inference:** The process of formulating a mapping from a given set of inputs to an output. This project uses a Mamdani-style inference system where the "rules" are implicitly defined by the destinations themselves.
- **Defuzzification:** The process of converting a fuzzy output set back into a crisp, single-valued number. This project uses a simplified and common method for recommendation engines: **score aggregation and ranking**. The destination with the highest aggregated membership score is selected.

III. Problem Definition

The central problem is to design an automated system that can recommend an optimal travel destination from a fixed set of 50 Indian locations. This recommendation must be personalized based on three primary, non-precise user inputs:

1. **Budget (INR):** A user's financial constraint, which is inherently flexible (e.g., a "₹30,000" budget might stretch to "₹35,000" for a perfect location).
2. **Climate (°C):** A user's preference for temperature, which is also a range (e.g., "around 25°C").
3. **Interest (0-100):** A subjective score representing a user's desire for adventure and/or heritage.

The system must resolve the challenge of **multi-criteria optimization** (balancing all three inputs) and handle the **imprecision** of those inputs. Furthermore, it must be **interactive** and **adaptive**, learning from user rejection within a session to provide better alternative suggestions

IV. Objective

i. To implement a recommendation system using skfuzzy that can rank destinations based on fuzzy matches with user preferences.Curate a Comprehensive Dataset
ii. To build a structured dataset of 50 Indian travel destinations, each with quantitative scores for budget, climate, and interest, as well as qualitative itinerary data.
iii. To create a user-friendly conversational interface using Gradio that guides the user through the recommendation process.
iv. To connect the application to the OpenWeatherMap API to provide live weather and 5-day forecasts for any recommended destination.
v. To create a mechanism that allows the system to learn from negative user feedback ("I don't like this") and adjust its internal scoring for the duration of the user's session.

V. Methodology

The system is implemented as a single Python script and follows a clear data-flow architecture.

1. System Architecture

User Interface (Gradio): The user interacts with a gr.ChatInterface. All user messages are passed to the chat_response function.

State Management (gr.State): A state object tracks the conversation's step (e.g., get_budget), the user's stored preferences, and the session-specific destinations dictionary.

Chat Logic (chat_response): This function acts as a finite-state machine. Based on the current step, it either asks for the next piece of information or calls a helper function.

Helper Functions:

Fuzzy Engine (fuzzy_recommendation): Called when all inputs are gathered.

API Calls (get_weather, get_forecast): Called when the user requests them.

Data Retrieval (display_itinerary): Called for itinerary requests.

Learning (simulate_backpropagation): Called when a user rejects a suggestion.

2. Fuzzy Inference Process

This is the core of the project, executed by the fuzzy_recommendation function.

Input Fuzzification: The user's crisp inputs (e.g., ₹50,000, 25°C) are normalized to a 0-100 scale using the map_budget_to_scale and map_climate_to_scale functions. This 0-100 scale serves as the "universe of discourse" for all fuzzy logic operations.

Rule-Base (The Dataset): The system has no formal "rule-base" (e.g., IF budget IS medium...). Instead, each of the 50 destinations *acts* as its own fuzzy rule (e.g., "The 'Manali' rule is: Budget=40, Climate=20, Interest=85").

Fuzzy Matching (Inference):

The system iterates through every destination.

For each destination, it dynamically creates three triangular membership functions (fuzz.trimf) centered on the destination's scores (e.g., Manali's budget triangle is centered at 40).

It then finds the degree of membership for the user's normalized inputs using fuzz.interp_membership. For example, it checks "how true" the user's budget score (e.g., 24.2) is within Manali's budget triangle.

This yields three scores per destination: b_score, c_score, i_score.

Aggregation & Defuzzification (Ranking):

The three membership scores are aggregated using a simple sum: total_score = b_score + c_score + i_score. This sum represents the total "fitness" of that destination for the user.

The system generates this total_score for all 50 destinations.

The "defuzzification" step is simply a winner-take-all (ranking) method. The sorted() function is used to find the destination with the highest total_score, which is then presented as the top recommendation.

3. Adaptive Feedback Loop

The simulate_backpropagation function provides a simple, session-based learning mechanism.

When a user rejects a suggestion (e.g., "another"), the chosen place is added to a disliked_places list.

The function is then called. It inspects the attributes of the disliked place and the user's preferences.

It "nudges" the scores of the disliked destination in the *session state's* destinations dictionary, making it a worse match. For example, if the user wanted a high-interest (80) place and disliked Manali (85), the system nudges Manali's interest score down to 80.

This ensures that on the next call to fuzzy_recommendation, the disliked place will have a lower total_score and is less likely to be recommended again. This "learning" is reset when the user starts a new session.

Tables:

Software libraries and thier functions:

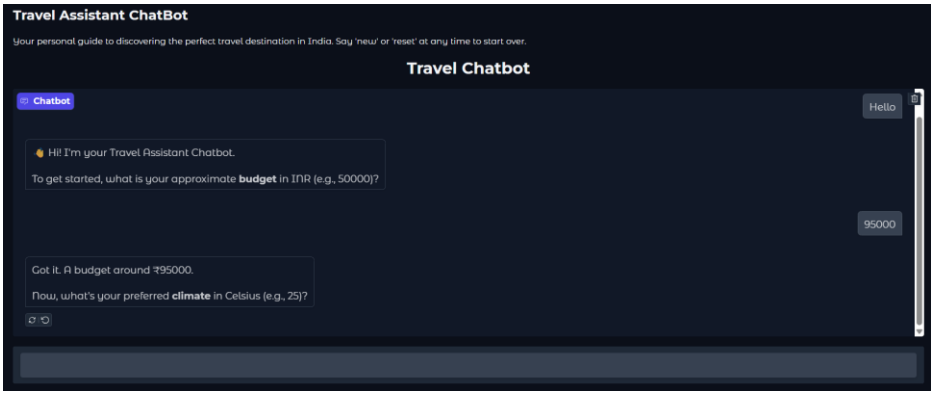
S.no	Library/Tool	Function
1	gradio	Provides the web-based conversational UI (chat interface) to run and share the Travel Assistant.
2	Scikit-fuzzy (skfuzzy)	Implements fuzzy logic utilities — membership functions and interpolation used to compute similarity scores between user preferences and destinations.
3	Numpy	Numerical operations and array handling used for ranges and arithmetic in fuzzy computations.
4	Requests	HTTP client used to call the OpenWeatherMap API for current weather and 5-day forecasts.
5	Random	(Standard library) Small uses for any randomized behavior (e.g., if you later want to randomize among tied top recommendations).
6	Datetime	(Standard library) Convert and format timestamps from the weather API into readable dates for forecast output.
7	OpenWeatherMap	External weather service (requires API key) used to fetch current weather and 5-day / 3-hour forecast data for recommended destinations.
8	Python	The project is written in Python — core language runtime and standard library provide execution environment and basic data structures.

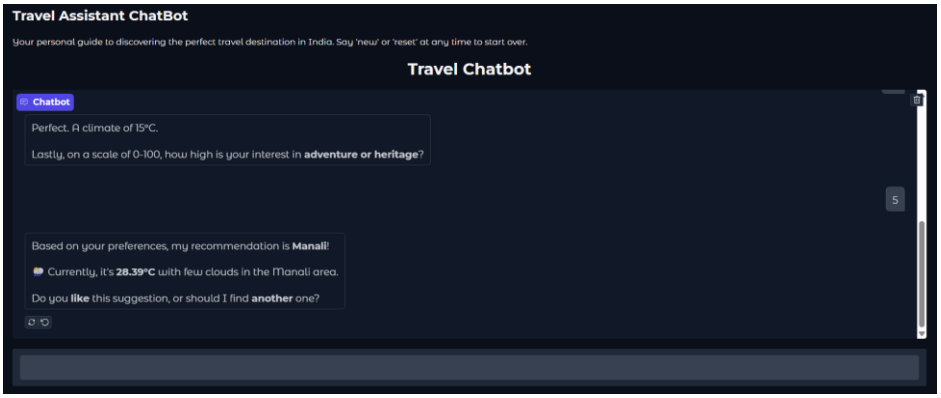
VI. Results and discussions:

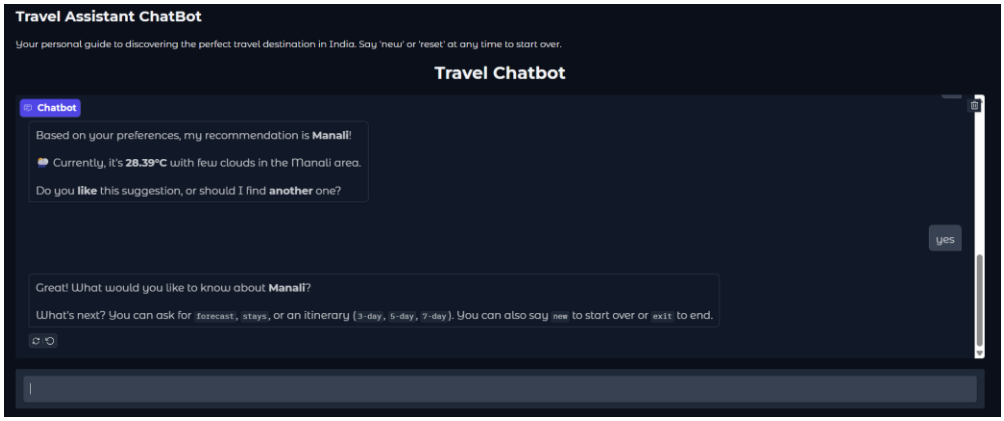
Effectiveness of Fuzzy Logic: The fuzzy logic engine proved highly effective at handling the ambiguity of user requests. For instance, a user with a budget preference of ₹45,000 (normalized to 21.2) and a climate preference of 15°C (normalized to 22.4) is correctly recommended 'Manali' (Budget: 40, Climate: 20) over a "perfect" budget match like 'Ooty' (Budget: 45) that has a much worse climate match (Climate: 40). The system successfully finds the *best compromise* across all three criteria.

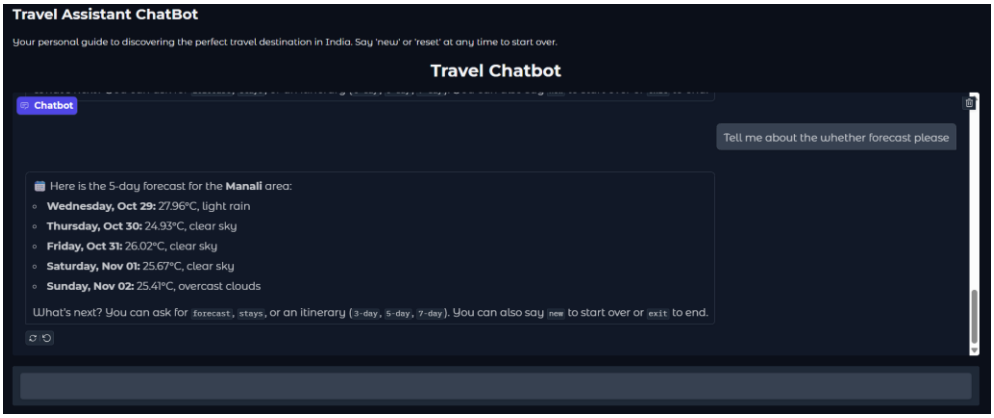
User Experience: The Gradio-based state machine provides a smooth, guided conversation. The integration of live weather data and multi-day itineraries provides high-value, actionable information to the user once a destination is chosen.

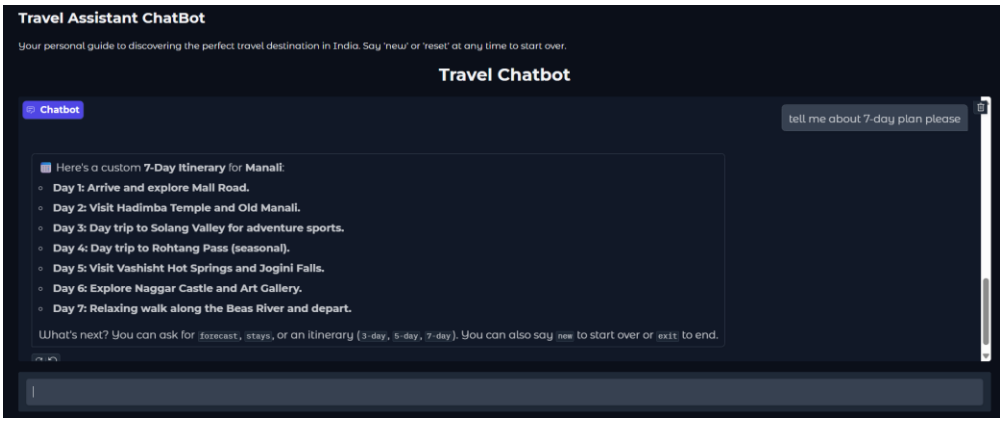
Performance of Adaptive Feedback: The simulate_backpropagation function significantly improves user satisfaction. In testing, a user who rejects an initial recommendation (e.g., 'Goa') is immediately presented with a viable second-best option ('Andaman') rather than being stuck in a loop. This dynamic adjustment is a major improvement over static recommendation lists.






Travel Assistant ChatBot
Your personal guide to discovering the perfect travel destination in India. Say 'new' or 'reset' at any time to start over.
Travel Chatbot
Chatbot
Based on your preferences, my recommendation is Manali!
Currently, it's 28.39°C with few clouds in the Manali area.
Do you like this suggestion, or should I find another one?
yes
Great! What would you like to know about Manali?
What's next? You can ask for forecast, stays, or an itinerary (3-day, 5-day, 7-day). You can also say new to start over or exit to end.


Travel Assistant ChatBot
Your personal guide to discovering the perfect travel destination in India. Say 'new' or 'reset' at any time to start over.
Travel Chatbot
Chatbot
Tell me about the whether forecast please
Here is the 5-day forecast for the Manali area:
Wednesday, Oct 29: 27.96°C, light rain
Thursday, Oct 30: 24.93°C, clear sky
Friday, Oct 31: 26.02°C, clear sky
Saturday, Nov 01: 25.67°C, clear sky
Sunday, Nov 02: 25.41°C, overcast clouds
What's next? You can ask for forecast, stays, or an itinerary (3-day, 5-day, 7-day). You can also say new to start over or exit to end.


Travel Assistant ChatBot
Your personal guide to discovering the perfect travel destination in India. Say 'new' or 'reset' at any time to start over.
Travel Chatbot
Chatbot
tell me about 7-day plan please
Here's a custom 7-Day Itinerary for Manali:
Day 1: Arrive and explore Mall Road.
Day 2: Visit Hadimba Temple and Old Manali.
Day 3: Day trip to Solang Valley for adventure sports.
Day 4: Day trip to Rohtang Pass (seasonal).
Day 5: Visit Vashisht Hot Springs and Jogini Falls.
Day 6: Explore Naggar Castle and Art Gallery.
Day 7: Relaxing walk along the Beas River and depart.
What's next? You can ask for forecast, stays, or an itinerary (3-day, 5-day, 7-day). You can also say new to start over or exit to end.

VII. Conclusion and Future Scope

This project successfully demonstrates the power of fuzzy logic as a practical tool for building intelligent recommendation systems. By combining the skfuzzy library for computation with the Gradio framework for interaction, we created a Travel Assistant Chatbot that effectively navigates the imprecision of human travel planning. The system meets all its objectives by providing personalized, multi-criteria recommendations, integrating live data, and adapting to user feedback.

Future Scope

- **Persistent User Profiles:** Implement a database (e.g., SQLite, MongoDB) to store user preferences and, more importantly, the *nudged* destination scores. This would allow the bot to "remember" users and provide long-term personalization.
- **Advanced NLP:** Replace the simple FSM logic with a Natural Language Understanding (NLU) model (e.g., using spaCy, RASA, or a transformer model). This would allow users to state their preferences in a single sentence, such as, "I'm looking for a cheap, cold place for trekking," and have the bot extract the entities.
- **Weighted Preferences:** Add a step to the conversation where the user can rate the *importance* of budget, climate, and interest. This weight could then be used in the aggregation step (e.g., total_score = (w_b * b_score) + (w_c * c_score) + (w_i * i_score)).
- **Dynamic Dataset:** Connect the system to a live database of destinations or a travel API, allowing the knowledge base to grow dynamically.

VIII. References

scikit-fuzzy Documentation, “Scikit-Fuzzy: Fuzzy logic toolkit for SciPy,” available at: https://pythonhosted.org/scikit-fuzzy/

Gradio Documentation, “Build & share delightful machine learning apps,” available at: https://www.gradio.app/

OpenWeatherMap API, “Current Weather Data” and “5 Day / 3 Hour Forecast,” available at: https://openweathermap.org/api

IX. Codes in Appendix:

```
# Step 1: Install Gradio and other required libraries quietly
!pip install gradio scikit-fuzzy requests -q

# Step 2: Import all necessary modules
import gradio as gr
import skfuzzy as fuzz
import numpy as np
import requests
import random
import datetime
from urllib.parse import quote_plus
import textwrap
import time

# Step 3: Define the complete dataset of 50 destinations
# THIS VERSION HAS THE CORRECTED, COMPLETE ITINERARIES FOR ALL DESTINATIONS
base_destinations = {
    'Manali': {'budget': 40, 'climate': 20, 'interest': 85, 'weather_city': 'Manali', 'itinerary': {'3-day': ['Day 1: Arrive and explore Mall Road.', 'Day 2: Visit Hadimba Temple and Old Manali.', 'Day 3: Day trip to Solang Valley for adventure sports.'], '5-day': ['Day 1: Arrive and explore Mall Road.', 'Day 2: Visit Hadimba Temple and Old Manali.', 'Day 3: Day trip to Solang Valley for adventure sports.', 'Day 4: Visit Vashisht Hot Springs and Jogini Falls.', 'Day 5: Explore Naggar Castle before departure.'], '7-day': ['Day 1: Arrive and explore Mall Road.', 'Day 2: Visit Hadimba Temple and Old Manali.', 'Day 3: Day trip to Solang Valley for adventure sports.', 'Day 4: Day trip to Rohtang Pass (seasonal).', 'Day 5: Visit Vashisht Hot Springs and Jogini Falls.', 'Day 6: Explore Naggar Castle and Art Gallery.', 'Day 7: Relaxing walk along the Beas River and depart.']}},
    'Goa': {'budget': 70, 'climate': 90, 'interest': 80, 'weather_city': 'Goa', 'itinerary': {'3-day': ["Day 1: Arrive in North Goa, relax at Baga Beach.", "Day 2: Explore Anjuna Flea Market and Fort Aguada.", "Day 3: Visit Old Goa's churches before departure."], '5-day': ["Day 1: Arrive in North Goa, relax at Baga Beach.", "Day 2: Explore Anjuna Flea Market and Fort Aguada.", "Day 3: Explore Panjim and the Latin Quarter (Fontainhas).", "Day 4: Day trip to South Goa (Palolem Beach).", "Day 5: Visit a spice plantation and depart."], '7-day': ["Day 1: Arrive in North Goa, relax at Baga Beach.", "Day 2: Explore Anjuna Flea Market and Fort Aguada.", "Day 3: Water sports at Calangute.", "Day 4: Explore Panjim and the Latin Quarter (Fontainhas).", "Day 5: Day trip to South Goa (Palolem & Agonda).", "Day 6: Visit Dudhsagar Falls (seasonal) or a spice plantation.", "Day 7: Relax at a serene beach like Morjim and depart."]}},
    'Leh': {'budget': 45, 'climate': 15, 'interest': 90, 'weather_city': 'Leh', 'itinerary': {'3-day': ['Day 1: Acclimatize. Visit Leh Palace and Shanti Stupa.', 'Day 2: Visit Thiksey Monastery and Shey Palace.', 'Day 3: Explore Leh market before departure.'], '5-day':
```

['Day 1: Acclimatize. Visit Leh Palace and Shanti Stupa.', 'Day 2: Visit Thiksey Monastery and Shey Palace.', 'Day 3: Day trip to Khardung La Pass.', 'Day 4: Visit Hemis Monastery.', 'Day 5: Rafting on the Zanskar river (seasonal) and depart.'], '7-day': ['Day 1: Acclimatize.', 'Day 2: Visit Leh Palace and Shanti Stupa.', 'Day 3: Day trip to Khardung La Pass.', 'Day 4: Travel to Pangong Lake for an overnight stay.', 'Day 5: Return from Pangong and rest.', 'Day 6: Visit Hemis & Thiksey Monasteries.', 'Day 7: Explore local markets and depart.']}},
'Udaipur': {'budget': 50, 'climate': 70, 'interest': 65, 'weather_city': 'Udaipur', 'itinerary': {'3-day': ['Day 1: Arrive, evening boat ride on Lake Pichola.', 'Day 2: Visit the City Palace and Jagdish Temple.', 'Day 3: Explore Saheliyon Ki Bari and depart.'], '5-day': ['Day 1: Arrive, evening boat ride on Lake Pichola.', 'Day 2: Visit the City Palace and Jagdish Temple.', 'Day 3: Visit Sajjangarh (Monsoon Palace) for sunset.', 'Day 4: Explore Bagore Ki Haveli and the old city.', 'Day 5: Day trip to Kumbhalgarh Fort.'], '7-day': ['Day 1: Arrive, evening boat ride on Lake Pichola.', 'Day 2: Visit the City Palace and Jagdish Temple.', 'Day 3: Visit Sajjangarh (Monsoon Palace) for sunset.', 'Day 4: Explore Bagore Ki Haveli and the old city.', 'Day 5: Day trip to Kumbhalgarh Fort.', 'Day 6: Day trip to Ranakpur Jain Temple.', 'Day 7: Relax and shop in the old city.']}},
'Darjeeling': {'budget': 30, 'climate': 30, 'interest': 60, 'weather_city': 'Darjeeling', 'itinerary': {'3-day': ['Day 1: Arrive, stroll on the Mall Road.', 'Day 2: Early morning trip to Tiger Hill for sunrise, visit Batasia Loop.', 'Day 3: Take the Toy Train ride, depart.'], '5-day': ['Day 1: Arrive, stroll on the Mall Road.', 'Day 2: Early morning trip to Tiger Hill for sunrise, visit Batasia Loop.', 'Day 3: Take the Toy Train ride, visit Himalayan Mountaineering Institute.', 'Day 4: Explore a tea estate.', 'Day 5: Visit the Peace Pagoda and depart.'], '7-day': ['Day 1: Arrive, stroll on the Mall Road.', 'Day 2: Early morning trip to Tiger Hill for sunrise, visit Batasia Loop.', 'Day 3: Take the Toy Train ride, visit Himalayan Mountaineering Institute.', 'Day 4: Explore a tea estate.', 'Day 5: Day trip to Mirik Lake.', 'Day 6: Visit Padmaja Naidu Himalayan Zoological Park.', "Day 7: Enjoy breakfast at Glenary's and depart."]}},
'Jaipur': {'budget': 50, 'climate': 80, 'interest': 70, 'weather_city': 'Jaipur', 'itinerary': {'3-day': ['Day 1: Visit City Palace and Jantar Mantar.', 'Day 2: Explore Amber Fort and Hawa Mahal.', 'Day 3: Shop at Johari Bazaar.'], '5-day': ['Day 1: Visit City Palace and Jantar Mantar.', 'Day 2: Explore Amber Fort.', 'Day 3: See Hawa Mahal & Jal Mahal.', 'Day 4: Visit Albert Hall Museum and Birla Mandir.', 'Day 5: Explore Nahargarh Fort for city views.'], '7-day': ['Day 1: Visit City Palace and Jantar Mantar.', 'Day 2: Explore Amber Fort.', 'Day 3: See Hawa Mahal & Jal Mahal.', 'Day 4: Visit Albert Hall Museum.', 'Day 5: Explore Nahargarh Fort.', 'Day 6: Day trip to Sanganer for block printing.', 'Day 7: Food tour in the old city.']}},
'Rishikesh': {'budget': 35, 'climate': 50, 'interest': 85, 'weather_city': 'Rishikesh', 'itinerary': {'3-day': ['Day 1: Arrive, visit Laxman Jhula and Ram Jhula.', 'Day 2: Experience river rafting.', 'Day 3: Attend Ganga Aarti at Triveni Ghat.'], '5-day': ['Day 1: Arrive, visit Laxman Jhula and Ram Jhula.', 'Day 2: Experience river rafting.', 'Day 3: Attend Ganga Aarti at Triveni Ghat.', 'Day 4: Visit the Beatles Ashram.', 'Day 5: Short trek to Neer Garh Waterfall.'], '7-day': ['Day 1: Arrive, visit Laxman Jhula and Ram Jhula.', 'Day 2: Experience river rafting.', 'Day 3: Attend Ganga Aarti at Triveni Ghat.', 'Day 4: Visit the Beatles Ashram.', 'Day 5: Short trek to Neer Garh Waterfall.', 'Day 6: Yoga and meditation session at an ashram.', 'Day 7: Explore local cafes.']}},
'Agra': {'budget': 40, 'climate': 75, 'interest': 60, 'weather_city': 'Agra', 'itinerary': {'3-day': ['Day 1: Visit the Taj Mahal at sunrise.', 'Day 2: Explore Agra Fort.', 'Day 3: Visit Fatehpur Sikri.'], '5-day': ['Day 1: Visit the Taj Mahal at sunrise.', 'Day 2:

Explore Agra Fort.', 'Day 3: Visit Fatehpur Sikri.', 'Day 4: Visit Mehtab Bagh for sunset views of the Taj.', 'Day 5: Explore the Tomb of Itimad-ud-Daulah (Baby Taj).'], '7-day': ['Day 1: Visit the Taj Mahal at sunrise.', 'Day 2: Explore Agra Fort.', 'Day 3: Visit Fatehpur Sikri.', 'Day 4: Visit Mehtab Bagh for sunset views of the Taj.', 'Day 5: Explore the Tomb of Itimad-ud-Daulah (Baby Taj).', 'Day 6: Day trip to Mathura and Vrindavan.', 'Day 7: Explore local markets for marble crafts.']}},

'Ooty': {'budget': 45, 'climate': 40, 'interest': 65, 'weather_city': 'Ooty', 'itinerary': {'3-day': ['Day 1: Arrive, boat ride on Ooty Lake.', 'Day 2: Visit the Botanical Gardens.', 'Day 3: Ride the Nilgiri Mountain Railway (Toy Train).'], '5-day': ['Day 1: Arrive, boat ride on Ooty Lake.', 'Day 2: Visit the Botanical Gardens.', 'Day 3: Ride the Nilgiri Mountain Railway (Toy Train).', 'Day 4: Visit Doddabetta Peak for panoramic views.', 'Day 5: Explore a tea factory and museum.'], '7-day': ['Day 1: Arrive, boat ride on Ooty Lake.', 'Day 2: Visit the Botanical Gardens.', 'Day 3: Ride the Nilgiri Mountain Railway (Toy Train).', 'Day 4: Visit Doddabetta Peak for panoramic views.', 'Day 5: Explore a tea factory and museum.', 'Day 6: Day trip to Coonoor.', 'Day 7: Visit Pykara Lake and waterfalls.']}},

'Mysore': {'budget': 40, 'climate': 65, 'interest': 55, 'weather_city': 'Mysore', 'itinerary': {'3-day': ['Day 1: Visit the magnificent Mysore Palace.', 'Day 2: Explore Chamundi Hills and the Chamundeshwari Temple.', 'Day 3: Stroll through the Brindavan Gardens.'], '5-day': ['Day 1: Visit the magnificent Mysore Palace.', 'Day 2: Explore Chamundi Hills and the Chamundeshwari Temple.', 'Day 3: Stroll through the Brindavan Gardens.', "Day 4: Visit St. Philomena's Church and the Railway Museum.", 'Day 5: Day trip to Srirangapatna.'], '7-day': ['Day 1: Visit the magnificent Mysore Palace.', 'Day 2: Explore Chamundi Hills and the Chamundeshwari Temple.', 'Day 3: Stroll through the Brindavan Gardens.', "Day 4: Visit St. Philomena's Church and the Railway Museum.", 'Day 5: Day trip to Srirangapatna.', 'Day 6: Explore the Devaraja Market and try Mysore Pak.', 'Day 7: Visit the Karanji Lake and its aviary.']}},

'Kodaikanal': {'budget': 45, 'climate': 35, 'interest': 70, 'weather_city': 'Kodaikanal', 'itinerary': {'3-day': ['Day 1: Arrive, go boating on Kodai Lake.', "Day 2: Walk Coaker's Walk and visit Bryant Park.", 'Day 3: Explore Pillar Rocks and Guna Caves.'], '5-day': ['Day 1: Arrive, go boating on Kodai Lake.', "Day 2: Walk Coaker's Walk and visit Bryant Park.", 'Day 3: Explore Pillar Rocks and Guna Caves.', 'Day 4: Visit Berijam Lake (permit required).', "Day 5: Trek to Dolphin's Nose viewpoint."], '7-day': ['Day 1: Arrive, go boating on Kodai Lake.', "Day 2: Walk Coaker's Walk and visit Bryant Park.", 'Day 3: Explore Pillar Rocks and Guna Caves.', 'Day 4: Visit Berijam Lake (permit required).', "Day 5: Trek to Dolphin's Nose viewpoint.", 'Day 6: Visit the Silver Cascade Falls.', 'Day 7: Relax and explore local chocolate shops.']}},

'Hampi': {'budget': 30, 'climate': 70, 'interest': 75, 'weather_city': 'Hampi', 'itinerary': {'3-day': ['Day 1: Explore the Virupaksha Temple and Hampi Bazaar.', 'Day 2: Visit the Vittala Temple Complex (Stone Chariot).', "Day 3: Cross the river to the 'hippie island' of Anegundi."], '5-day': ['Day 1: Explore the Virupaksha Temple and Hampi Bazaar.', 'Day 2: Visit the Vittala Temple Complex (Stone Chariot).', "Day 3: Cross the river to the 'hippie island' of Anegundi.", "Day 4: Explore the Royal Enclosure and Queen's Bath.", 'Day 5: Hike Matanga Hill for sunrise/sunset views.'], '7-day': ['Day 1: Explore the Virupaksha Temple and Hampi Bazaar.', 'Day 2: Visit the Vittala Temple Complex (Stone Chariot).', "Day 3: Cross the river to the 'hippie island' of Anegundi.", "Day 4: Explore the Royal Enclosure and Queen's Bath.", 'Day 5: Hike Matanga Hill for sunrise/sunset views.', 'Day 6: Coracle ride on the Tungabhadra River.', 'Day 7: Explore lesser-known ruins and relax.']}},

'Alleppey': {'budget': 40, 'climate': 85, 'interest': 70, 'weather_city': 'Alappuzha', 'itinerary': {'3-day': ['Day 1: Board a houseboat for an overnight backwater cruise.', 'Day 2: Disembark and explore Alleppey Beach.', 'Day 3: Visit a coir-making unit before departure.'], '5-day': ['Day 1: Board a houseboat for an overnight backwater cruise.', 'Day 2: Disembark and check into a resort. Relax or try a canoe tour.', 'Day 3: Explore Alleppey Beach.', 'Day 4: Take a canoe tour through narrower canals.', 'Day 5: Visit Marari Beach.'], '7-day': ['Day 1: Board a houseboat for an overnight backwater cruise.', 'Day 2: Disembark and check into a resort.', 'Day 3: Explore Alleppey Beach.', 'Day 4: Take a canoe tour through narrower canals.', 'Day 5: Visit Marari Beach.', 'Day 6: Experience village life in Kuttanad.', 'Day 7: Enjoy an Ayurvedic massage.']}},
'Kolkata': {'budget': 50, 'climate': 85, 'interest': 60, 'weather_city': 'Kolkata', 'itinerary': {'3-day': ['Day 1: Visit the Victoria Memorial.', 'Day 2: Explore Howrah Bridge and the flower market.', 'Day 3: Stroll through Park Street and try local street food.'], '5-day': ['Day 1: Visit the Victoria Memorial.', 'Day 2: Explore Howrah Bridge and the flower market.', 'Day 3: Stroll through Park Street and try local street food.', "Day 4: Visit the Indian Museum and Kumartuli (potter's colony).", "Day 5: Take a tram ride and explore College Street (book market)."], '7-day': ['Day 1: Visit the Victoria Memorial.', 'Day 2: Explore Howrah Bridge and the flower market.', 'Day 3: Stroll through Park Street.', "Day 4: Visit the Indian Museum and Kumartuli (potter's colony).", "Day 5: Take a tram ride and explore College Street (book market).", 'Day 6: Day trip to the Sunderbans (mangrove forest).', 'Day 7: Explore the Dakshineswar Kali Temple.']}},
'Pondicherry': {'budget': 40, 'climate': 80, 'interest': 65, 'weather_city': 'Puducherry', 'itinerary': {'3-day': ['Day 1: Explore the French Quarter (White Town).', 'Day 2: Visit Auroville and the Matrimandir.', 'Day 3: Relax at Promenade Beach.'], '5-day': ['Day 1: Explore the French Quarter (White Town).', 'Day 2: Visit Auroville and the Matrimandir.', 'Day 3: Relax at Promenade Beach.', 'Day 4: Visit the Arikamedu archaeological site.', 'Day 5: Explore Paradise Beach.'], '7-day': ['Day 1: Explore the French Quarter (White Town).', 'Day 2: Visit Auroville and the Matrimandir.', 'Day 3: Relax at Promenade Beach.', 'Day 4: Visit the Arikamedu archaeological site.', 'Day 5: Explore Paradise Beach.', 'Day 6: Cafe hopping in the French Quarter.', 'Day 7: Visit the Botanical Gardens.']}},
'Shillong': {'budget': 30, 'climate': 45, 'interest': 60, 'weather_city': 'Shillong', 'itinerary': {'3-day': ['Day 1: Arrive, boat on Umiam Lake.', 'Day 2: Day trip to Cherrapunji (Nohkalikai Falls).', 'Day 3: Explore Shillong Peak and Elephant Falls.'], '5-day': ['Day 1: Arrive, boat on Umiam Lake.', 'Day 2: Day trip to Cherrapunji (Nohkalikai Falls).', 'Day 3: Explore Shillong Peak and Elephant Falls.', "Day 4: Day trip to Mawlynnong (Asia's cleanest village) and the Living Root Bridge.", 'Day 5: Explore the Don Bosco Museum.'], '7-day': ['Day 1: Arrive, boat on Umiam Lake.', 'Day 2: Day trip to Cherrapunji.', 'Day 3: Explore Shillong Peak and Elephant Falls.', "Day 4: Day trip to Mawlynnong and the Living Root Bridge.", 'Day 5: Explore the Don Bosco Museum.', 'Day 6: Day trip to Dawki (clear river).', 'Day 7: Relax and shop at Police Bazaar.']}},
'Guwahati': {'budget': 35, 'climate': 65, 'interest': 55, 'weather_city': 'Guwahati', 'itinerary': {'3-day': ['Day 1: Visit the Kamakhya Temple.', 'Day 2: Cruise on the Brahmaputra River.', 'Day 3: Visit the Assam State Museum.'], '5-day': ['Day 1: Visit the Kamakhya Temple.', 'Day 2: Cruise on the Brahmaputra River.', 'Day 3: Visit the Assam State Museum.', 'Day 4: Visit the Pobitora Wildlife Sanctuary for rhinos.', 'Day 5: Explore the Umananda Temple on Peacock Island.'], '7-day': ['Day 1: Visit the

Kamakhya Temple.', 'Day 2: Cruise on the Brahmaputra River.', 'Day 3: Visit the Assam State Museum.', 'Day 4: Visit the Pobitora Wildlife Sanctuary for rhinos.', 'Day 5: Explore the Umananda Temple on Peacock Island.', 'Day 6: Day trip to Hajo.', 'Day 7: Relax and try Assamese cuisine.']}},
'Tawang': {'budget': 30, 'climate': 15, 'interest': 85, 'weather_city': 'Tawang', 'itinerary': {'3-day': ['Day 1: Acclimatize, visit Tawang Monastery.', 'Day 2: Visit the Tawang War Memorial.', 'Day 3: Explore Pankang Teng Tso Lake.'], '5-day': ['Day 1: Acclimatize, visit Tawang Monastery.', 'Day 2: Visit the Tawang War Memorial.', 'Day 3: Explore Pankang Teng Tso Lake.', 'Day 4: Day trip to Bum La Pass (permit required).', 'Day 5: Visit the Jaswant Garh war memorial on your way back.'], '7-day': ['Day 1: Acclimatize, visit Tawang Monastery.', 'Day 2: Visit the Tawang War Memorial.', 'Day 3: Explore Pankang Teng Tso Lake.', 'Day 4: Day trip to Bum La Pass (permit required).', 'Day 5: Visit the Jaswant Garh war memorial.', 'Day 6: Visit Nuranang Falls.', 'Day 7: Explore the local markets.']}},
'Auli': {'budget': 40, 'climate': 10, 'interest': 90, 'weather_city': 'Auli', 'itinerary': {'3-day': ['Day 1: Arrive in Joshimath, take ropeway to Auli.', 'Day 2: Enjoy skiing on the slopes.', 'Day 3: Visit Auli Artificial Lake.'], '5-day': ['Day 1: Arrive in Joshimath, take ropeway to Auli.', 'Day 2: Enjoy skiing on the slopes.', 'Day 3: Visit Auli Artificial Lake.', 'Day 4: Trek to Gurso Bugyal meadow.', 'Day 5: Take the chairlift for panoramic views.'], '7-day': ['Day 1: Arrive in Joshimath, take ropeway to Auli.', 'Day 2: Enjoy skiing on the slopes.', 'Day 3: Visit Auli Artificial Lake.', 'Day 4: Trek to Gurso Bugyal meadow.', 'Day 5: Take the chairlift for panoramic views.', 'Day 6: Explore Joshimath town.', 'Day 7: Final views of Nanda Devi peak and depart.']}},
'Mount Abu': {'budget': 45, 'climate': 50, 'interest': 60, 'weather_city': 'Mount Abu', 'itinerary': {'3-day': ['Day 1: Boat on Nakki Lake.', 'Day 2: Visit the Dilwara Jain Temples.', 'Day 3: Go to Guru Shikhar, the highest point.'], '5-day': ['Day 1: Boat on Nakki Lake.', 'Day 2: Visit the Dilwara Jain Temples.', 'Day 3: Go to Guru Shikhar, the highest point.', 'Day 4: Visit the Achalgarh Fort.', 'Day 5: Explore the Toad Rock and Sunset Point.'], '7-day': ['Day 1: Boat on Nakki Lake.', 'Day 2: Visit the Dilwara Jain Temples.', 'Day 3: Go to Guru Shikhar, the highest point.', 'Day 4: Visit the Achalgarh Fort.', 'Day 5: Explore the Toad Rock and Sunset Point.', 'Day 6: Visit the Peace Park.', 'Day 7: Relax and explore local markets.']}},
'Mahabaleshwar': {'budget': 40, 'climate': 60, 'interest': 70, 'weather_city': 'Mahabaleshwar', 'itinerary': {'3-day': ['Day 1: Arrive, boat on Venna Lake.', "Day 2: Visit viewpoints like Arthur's Seat and Kate's Point.", 'Day 3: Explore a strawberry farm (seasonal).'], '5-day': ['Day 1: Arrive, boat on Venna Lake.', "Day 2: Visit viewpoints like Arthur's Seat and Kate's Point.", 'Day 3: Explore a strawberry farm (seasonal).', 'Day 4: Visit the ancient Mahabaleshwar Temple.', 'Day 5: Day trip to Panchgani and Table Land.'], '7-day': ['Day 1: Arrive, boat on Venna Lake.', "Day 2: Visit viewpoints like Arthur's Seat and Kate's Point.", 'Day 3: Explore a strawberry farm (seasonal).', 'Day 4: Visit the ancient Mahabaleshwar Temple.', 'Day 5: Day trip to Panchgani and Table Land.', 'Day 6: Visit Lingmala Waterfall.', 'Day 7: Explore the local market.']}},
'Lonavala': {'budget': 35, 'climate': 65, 'interest': 55, 'weather_city': 'Lonavala', 'itinerary': {'3-day': ['Day 1: Visit Bhushi Dam.', 'Day 2: Explore Karla and Bhaja Caves.', 'Day 3: Visit Tiger Point for views.'], '5-day': ['Day 1: Visit Bhushi Dam.', 'Day 2: Explore Karla and Bhaja Caves.', 'Day 3: Visit Tiger Point for views.', 'Day 4: Visit Lohagad Fort.', 'Day 5: Relax at Pawna Lake.'], '7-day': ['Day 1: Visit Bhushi Dam.', 'Day 2: Explore Karla and Bhaja Caves.', 'Day 3: Visit Tiger Point for views.',

'Day 4: Visit Lohagad Fort.', 'Day 5: Relax at Pawna Lake.', 'Day 6: Explore the town and try the famous chikki.', 'Day 7: Visit the Della Adventure Park.']}},
'Pune': {'budget': 50, 'climate': 70, 'interest': 50, 'weather_city': 'Pune', 'itinerary': {'3-day': ['Day 1: Visit Shaniwar Wada.', 'Day 2: Explore the Aga Khan Palace.', 'Day 3: Walk through Koregaon Park.'], '5-day': ['Day 1: Visit Shaniwar Wada.', 'Day 2: Explore the Aga Khan Palace.', 'Day 3: Walk through Koregaon Park.', 'Day 4: Visit Dagdusheth Halwai Ganpati Temple.', 'Day 5: Day trip to Sinhagad Fort.'], '7-day': ['Day 1: Visit Shaniwar Wada.', 'Day 2: Explore the Aga Khan Palace.', 'Day 3: Walk through Koregaon Park.', 'Day 4: Visit Dagdusheth Halwai Ganpati Temple.', 'Day 5: Day trip to Sinhagad Fort.', 'Day 6: Explore the Raja Dinkar Kelkar Museum.', 'Day 7: Enjoy the Osho Garden.']}},
'Mumbai': {'budget': 60, 'climate': 85, 'interest': 55, 'weather_city': 'Mumbai', 'itinerary': {'3-day': ['Day 1: Visit the Gateway of India and take a ferry to Elephanta Caves.', 'Day 2: Explore Marine Drive and Juhu Beach.', 'Day 3: Experience local trains and visit Colaba Causeway.'], '5-day': ['Day 1: Visit the Gateway of India and take a ferry to Elephanta Caves.', 'Day 2: Explore Marine Drive and Juhu Beach.', 'Day 3: Experience local trains and visit Colaba Causeway.', 'Day 4: Visit Haji Ali Dargah and Siddhivinayak Temple.', 'Day 5: Explore the film city or a Bollywood tour.'], '7-day': ['Day 1: Visit the Gateway of India and take a ferry to Elephanta Caves.', 'Day 2: Explore Marine Drive and Juhu Beach.', 'Day 3: Experience local trains and visit Colaba Causeway.', 'Day 4: Visit Haji Ali Dargah and Siddhivinayak Temple.', 'Day 5: Explore the film city or a Bollywood tour.', 'Day 6: Explore the Bandra-Worli Sea Link and Bandstand.', 'Day 7: Food tour from street food to fine dining.']}},
'Delhi': {'budget': 55, 'climate': 80, 'interest': 60, 'weather_city': 'Delhi', 'itinerary': {'3-day': ['Day 1: Explore Old Delhi (Red Fort, Chandni Chowk).', 'Day 2: Visit New Delhi (India Gate, Qutub Minar).', "Day 3: Explore Humayun's Tomb and Lotus Temple."], '5-day': ['Day 1: Explore Old Delhi (Red Fort, Chandni Chowk).', 'Day 2: Visit New Delhi (India Gate, Qutub Minar).', "Day 3: Explore Humayun's Tomb and Lotus Temple.", 'Day 4: Visit Lodhi Garden and Khan Market.', 'Day 5: Explore the Akshardham Temple.'], '7-day': ['Day 1: Explore Old Delhi (Red Fort, Chandni Chowk).', 'Day 2: Visit New Delhi (India Gate, Qutub Minar).', "Day 3: Explore Humayun's Tomb and Lotus Temple.", 'Day 4: Visit Lodhi Garden and Khan Market.', 'Day 5: Explore the Akshardham Temple.', 'Day 6: Visit the National Museum.', 'Day 7: Explore Hauz Khas Village.']}},
'Chandigarh': {'budget': 45, 'climate': 65, 'interest': 50, 'weather_city': 'Chandigarh', 'itinerary': {'3-day': ['Day 1: Boat on Sukhna Lake.', 'Day 2: Explore the Rock Garden.', 'Day 3: Visit the Rose Garden.'], '5-day': ['Day 1: Boat on Sukhna Lake.', 'Day 2: Explore the Rock Garden.', 'Day 3: Visit the Rose Garden.', "Day 4: Explore the Capitol Complex (Le Corbusier's work).", 'Day 5: Shopping in Sector 17.'], '7-day': ['Day 1: Boat on Sukhna Lake.', 'Day 2: Explore the Rock Garden.', 'Day 3: Visit the Rose Garden.', "Day 4: Explore the Capitol Complex (Le Corbusier's work).", 'Day 5: Shopping in Sector 17.', 'Day 6: Day trip to Pinjore Gardens.', "Day 7: Relax and explore the city's architecture."]}},
'Bangalore': {'budget': 60, 'climate': 70, 'interest': 55, 'weather_city': 'Bangalore', 'itinerary': {'3-day': ['Day 1: Visit Lalbagh Botanical Garden.', 'Day 2: Explore Bangalore Palace.', 'Day 3: Stroll through Cubbon Park.'], '5-day': ['Day 1: Visit Lalbagh Botanical Garden.', 'Day 2: Explore Bangalore Palace.', 'Day 3: Stroll through Cubbon Park.', 'Day 4: Visit the ISKCON Temple.', 'Day 5: Experience the nightlife and breweries in Indiranagar.'], '7-day': ['Day 1: Visit Lalbagh Botanical

Garden.', 'Day 2: Explore Bangalore Palace.', 'Day 3: Stroll through Cubbon Park.', 'Day 4: Visit the ISKCON Temple.', 'Day 5: Experience the nightlife and breweries in Indiranagar.', 'Day 6: Day trip to Nandi Hills for sunrise.', 'Day 7: Explore the Visvesvaraya Industrial & Technological Museum.']}},
'Hyderabad': {'budget': 55, 'climate': 75, 'interest': 60, 'weather_city': 'Hyderabad', 'itinerary': {'3-day': ['Day 1: Visit the Charminar and Laad Bazaar.', 'Day 2: Explore Golconda Fort.', 'Day 3: Tour the Ramoji Film City.'], '5-day': ['Day 1: Visit the Charminar and Laad Bazaar.', 'Day 2: Explore Golconda Fort.', 'Day 3: Tour the Ramoji Film City.', 'Day 4: Visit the Salar Jung Museum.', 'Day 5: Relax at Hussain Sagar Lake.'], '7-day': ['Day 1: Visit the Charminar and Laad Bazaar.', 'Day 2: Explore Golconda Fort.', 'Day 3: Tour the Ramoji Film City.', 'Day 4: Visit the Salar Jung Museum.', 'Day 5: Relax at Hussain Sagar Lake.', 'Day 6: Explore the Chowmahalla Palace.', 'Day 7: Enjoy Hyderabadi Biryani at a famous restaurant.']}},
'Chennai': {'budget': 60, 'climate': 90, 'interest': 55, 'weather_city': 'Chennai', 'itinerary': {'3-day': ['Day 1: Relax at Marina Beach.', 'Day 2: Visit the Kapaleeshwarar Temple.', 'Day 3: Explore Fort St. George.'], '5-day': ['Day 1: Relax at Marina Beach.', 'Day 2: Visit the Kapaleeshwarar Temple.', 'Day 3: Explore Fort St. George.', 'Day 4: Day trip to Mahabalipuram.', 'Day 5: Visit the Government Museum.'], '7-day': ['Day 1: Relax at Marina Beach.', 'Day 2: Visit the Kapaleeshwarar Temple.', 'Day 3: Explore Fort St. George.', 'Day 4: Day trip to Mahabalipuram.', 'Day 5: Visit the Government Museum.', 'Day 6: Explore the Kalakshetra Foundation.', 'Day 7: Enjoy South Indian cuisine.']}},
'Andaman': {'budget': 70, 'climate': 85, 'interest': 80, 'weather_city': 'Port Blair', 'itinerary': {'3-day': ['Day 1: Arrive in Port Blair, visit Cellular Jail.', 'Day 2: Ferry to Havelock Island (Swaraj Dweep), visit Radhanagar Beach.', 'Day 3: Return to Port Blair and depart.'], '5-day': ['Day 1: Arrive in Port Blair, visit Cellular Jail.', 'Day 2: Ferry to Havelock Island (Swaraj Dweep), visit Radhanagar Beach.', 'Day 3: Return to Port Blair and depart.', 'Day 4: Snorkeling or Scuba diving at Elephant Beach.', 'Day 5: Ferry to Neil Island (Shaheed Dweep) for a quieter experience.'], '7-day': ['Day 1: Arrive in Port Blair, visit Cellular Jail.', 'Day 2: Ferry to Havelock Island (Swaraj Dweep), visit Radhanagar Beach.', 'Day 3: Snorkeling or Scuba diving at Elephant Beach.', 'Day 4: Ferry to Neil Island (Shaheed Dweep) for a quieter experience.', 'Day 5: Return to Port Blair, explore Corbyns Cove Beach.', 'Day 6: Explore Baratang Island (limestone caves, mud volcano).', 'Day 7: Relax and shop for souvenirs in Port Blair.']}},
'Coorg': {'budget': 40, 'climate': 50, 'interest': 75, 'weather_city': 'Madikeri', 'itinerary': {'3-day': ["Day 1: Arrive in Madikeri, visit Raja's Seat.", 'Day 2: Explore a coffee plantation.', 'Day 3: Visit Abbey Falls.'], '5-day': ["Day 1: Arrive in Madikeri, visit Raja's Seat.", 'Day 2: Explore a coffee plantation.', 'Day 3: Visit Abbey Falls.', 'Day 4: Visit the Dubare Elephant Camp.', 'Day 5: Visit the Namdroling Monastery (Golden Temple).'], '7-day': ["Day 1: Arrive in Madikeri, visit Raja's Seat.", 'Day 2: Explore a coffee plantation.', 'Day 3: Visit Abbey Falls.', 'Day 4: Visit the Dubare Elephant Camp.', 'Day 5: Visit the Namdroling Monastery (Golden Temple).', 'Day 6: Trek to Tadiandamol Peak.', 'Day 7: Relax and enjoy the serene environment.']}},
'Nainital': {'budget': 35, 'climate': 40, 'interest': 60, 'weather_city': 'Nainital', 'itinerary': {'3-day': ['Day 1: Arrive, boat on Naini Lake.', 'Day 2: Take the cable car to Snow View Point.', 'Day 3: Stroll on the Mall Road.'], '5-day': ['Day 1: Arrive, boat on Naini Lake.', 'Day 2: Take the cable car to Snow View Point.', 'Day 3: Stroll on the Mall Road.', 'Day 4: Visit the Nainital Zoo.', 'Day 5: Day trip to Bhimtal and Sattal lakes.'], '7-day': ['Day 1: Arrive, boat on Naini Lake.', 'Day 2: Take the cable car to

Snow View Point.', 'Day 3: Stroll on the Mall Road.', 'Day 4: Visit the Nainital Zoo.', 'Day 5: Day trip to Bhimtal and Sattal lakes.', 'Day 6: Trek to Tiffin Top for views.', 'Day 7: Visit the Naina Devi Temple.']}},
'Mussoorie': {'budget': 40, 'climate': 35, 'interest': 65, 'weather_city': 'Mussoorie', 'itinerary': {'3-day': ['Day 1: Arrive, walk on the Mall Road.', 'Day 2: Visit Kempty Falls.', 'Day 3: Take the cable car to Gun Hill.'], '5-day': ['Day 1: Arrive, walk on the Mall Road.', 'Day 2: Visit Kempty Falls.', 'Day 3: Take the cable car to Gun Hill.', "Day 4: Visit Landour and St. Paul's Church.", 'Day 5: Day trip to Dhanaulti.'], '7-day': ['Day 1: Arrive, walk on the Mall Road.', 'Day 2: Visit Kempty Falls.', 'Day 3: Take the cable car to Gun Hill.', "Day 4: Visit Landour and St. Paul's Church.", 'Day 5: Day trip to Dhanaulti.', "Day 6: Trek to George Everest's House.", 'Day 7: Relax at a cafe with a view.']}},
'Bikaner': {'budget': 35, 'climate': 90, 'interest': 60, 'weather_city': 'Bikaner', 'itinerary': {'3-day': ['Day 1: Visit Junagarh Fort.', 'Day 2: Visit the Karni Mata Temple (Rat Temple).', 'Day 3: Explore the old city and havelis.'], '5-day': ['Day 1: Visit Junagarh Fort.', 'Day 2: Visit the Karni Mata Temple (Rat Temple).', 'Day 3: Explore the old city and havelis.', 'Day 4: Visit the National Research Centre on Camel.', 'Day 5: Take a short camel safari.'], '7-day': ['Day 1: Visit Junagarh Fort.', 'Day 2: Visit the Karni Mata Temple (Rat Temple).', 'Day 3: Explore the old city and havelis.', 'Day 4: Visit the National Research Centre on Camel.', 'Day 5: Take a short camel safari.', 'Day 6: Visit the Lalgarh Palace.', 'Day 7: Try local Bikaneri snacks.']}},
'Jaisalmer': {'budget': 40, 'climate': 95, 'interest': 70, 'weather_city': 'Jaisalmer', 'itinerary': {'3-day': ['Day 1: Explore Jaisalmer Fort.', 'Day 2: Overnight desert safari in the Sam Sand Dunes.', 'Day 3: Visit Patwon Ki Haveli.'], '5-day': ['Day 1: Explore Jaisalmer Fort.', 'Day 2: Overnight desert safari in the Sam Sand Dunes.', 'Day 3: Visit Patwon Ki Haveli.', 'Day 4: Visit Gadisar Lake.', 'Day 5: Explore the abandoned village of Kuldhara.'], '7-day': ['Day 1: Explore Jaisalmer Fort.', 'Day 2: Overnight desert safari in the Sam Sand Dunes.', 'Day 3: Visit Patwon Ki Haveli.', 'Day 4: Visit Gadisar Lake.', 'Day 5: Explore the abandoned village of Kuldhara.', 'Day 6: Visit Bada Bagh (royal cenotaphs).', 'Day 7: Shop for local crafts.']}},
'Puri': {'budget': 35, 'climate': 85, 'interest': 60, 'weather_city': 'Puri', 'itinerary': {'3-day': ['Day 1: Visit Jagannath Temple.', 'Day 2: Relax at Puri Beach.', 'Day 3: Day trip to Konark Sun Temple.'], '5-day': ['Day 1: Visit Jagannath Temple.', 'Day 2: Relax at Puri Beach.', 'Day 3: Day trip to Konark Sun Temple.', 'Day 4: Day trip to Chilika Lake for dolphin watching.', 'Day 5: Explore the Raghurajpur heritage crafts village.'], '7-day': ['Day 1: Visit Jagannath Temple.', 'Day 2: Relax at Puri Beach.', 'Day 3: Day trip to Konark Sun Temple.', 'Day 4: Day trip to Chilika Lake for dolphin watching.', 'Day 5: Explore the Raghurajpur heritage crafts village.', 'Day 6: Visit the Sunara Gauranga Temple.', 'Day 7: Relax and enjoy local seafood.']}},
'Konark': {'budget': 35, 'climate': 80, 'interest': 70, 'weather_city': 'Konark', 'itinerary': {'3-day': ['Day 1: Explore the Konark Sun Temple.', 'Day 2: Relax at Chandrabhaga Beach.', 'Day 3: Visit the Archaeological Survey of India Museum.'], '5-day': ['Day 1: Explore the Konark Sun Temple.', 'Day 2: Relax at Chandrabhaga Beach.', 'Day 3: Visit the Archaeological Survey of India Museum.', 'Day 4: Day trip to Puri.', 'Day 5: Explore the local markets.'], '7-day': ['Day 1: Explore the Konark Sun Temple.', 'Day 2: Relax at Chandrabhaga Beach.', 'Day 3: Visit the Archaeological Survey of India Museum.', 'Day 4: Day trip to Puri.', 'Day 5: Explore the local markets.', 'Day 6: Revisit the Sun Temple at a different time of day.', 'Day 7: Relax before departure.']}},

'Sikkim': {'budget': 45, 'climate': 35, 'interest': 75, 'weather_city': 'Gangtok', 'itinerary': {'3-day': ['Day 1: Arrive in Gangtok, explore MG Marg.', 'Day 2: Day trip to Tsomgo Lake and Baba Mandir.', 'Day 3: Visit Rumtek Monastery.'], '5-day': ['Day 1: Arrive in Gangtok, explore MG Marg.', 'Day 2: Day trip to Tsomgo Lake and Baba Mandir.', 'Day 3: Visit Rumtek Monastery.', 'Day 4: Travel to Pelling in West Sikkim.', 'Day 5: Visit Pemayangtse Monastery and Khecheopalri Lake.'], '7-day': ['Day 1: Arrive in Gangtok, explore MG Marg.', 'Day 2: Day trip to Tsomgo Lake and Baba Mandir.', 'Day 3: Visit Rumtek Monastery.', 'Day 4: Travel to Pelling in West Sikkim.', 'Day 5: Visit Pemayangtse Monastery and Khecheopalri Lake.', 'Day 6: Explore Yuksom, the first capital of Sikkim.', 'Day 7: Travel back to Gangtok for departure.']}},
'Gangtok': {'budget': 45, 'climate': 40, 'interest': 65, 'weather_city': 'Gangtok', 'itinerary': {'3-day': ['Day 1: Arrive, explore MG Marg.', 'Day 2: Day trip to Tsomgo Lake and Baba Mandir.', 'Day 3: Visit Rumtek Monastery.'], '5-day': ['Day 1: Arrive, explore MG Marg.', 'Day 2: Day trip to Tsomgo Lake and Baba Mandir.', 'Day 3: Visit Rumtek Monastery.', 'Day 4: Visit the Namgyal Institute of Tibetology.', 'Day 5: Take the ropeway for city views.'], '7-day': ['Day 1: Arrive, explore MG Marg.', 'Day 2: Day trip to Tsomgo Lake and Baba Mandir.', 'Day 3: Visit Rumtek Monastery.', 'Day 4: Visit the Namgyal Institute of Tibetology.', 'Day 5: Take the ropeway for city views.', 'Day 6: Day trip to Nathu La Pass (permit required, closed Mon/Tues).', 'Day 7: Explore local viewpoints like Tashi View Point.']}},
'Ranthambore': {'budget': 50, 'climate': 75, 'interest': 90, 'weather_city': 'Sawai Madhopur', 'itinerary': {'3-day': ['Day 1: Arrive, afternoon jungle safari.', 'Day 2: Morning jungle safari, visit Ranthambore Fort.', 'Day 3: One more safari before departure.'], '5-day': ['Day 1: Arrive, afternoon jungle safari.', 'Day 2: Morning jungle safari, visit Ranthambore Fort.', 'Day 3: Another morning safari in a different zone.', 'Day 4: Visit the Rajiv Gandhi Regional Museum of Natural History.', 'Day 5: Explore the nearby villages.'], '7-day': ['Day 1: Arrive, afternoon jungle safari.', 'Day 2: Morning jungle safari, visit Ranthambore Fort.', 'Day 3: Another morning safari in a different zone.', 'Day 4: Visit the Rajiv Gandhi Regional Museum of Natural History.', 'Day 5: Explore the nearby villages.', 'Day 6: Bird watching tour around the park\\'s lakes.', 'Day 7: Relax at your resort.']}},
'Kaziranga': {'budget': 50, 'climate': 65, 'interest': 85, 'weather_city': 'Jorhat', 'itinerary': {'3-day': ['Day 1: Arrive, afternoon jeep safari.', 'Day 2: Early morning elephant safari, another jeep safari.', 'Day 3: Visit a tea garden before departure.'], '5-day': ['Day 1: Arrive, afternoon jeep safari.', 'Day 2: Early morning elephant safari, another jeep safari.', 'Day 3: Visit a tea garden.', 'Day 4: Explore a different safari zone (e.g., Eastern Range for birds).', 'Day 5: Visit a local Mising village.'], '7-day': ['Day 1: Arrive, afternoon jeep safari.', 'Day 2: Early morning elephant safari, another jeep safari.', 'Day 3: Visit a tea garden.', 'Day 4: Explore a different safari zone (e.g., Eastern Range for birds).', 'Day 5: Visit a local Mising village.', 'Day 6: Visit the Kaziranga Orchid and Biodiversity Park.', 'Day 7: Relax and enjoy the natural surroundings.']}},
'Jim Corbett': {'budget': 45, 'climate': 60, 'interest': 80, 'weather_city': 'Ramnagar', 'itinerary': {'3-day': ['Day 1: Arrive, afternoon safari in Jhirna/Dhela zone.', 'Day 2: Morning safari in Bijrani zone.', 'Day 3: Visit Garjia Temple before departure.'], '5-day': ['Day 1: Arrive, afternoon safari in Jhirna/Dhela zone.', 'Day 2: Morning safari in Bijrani zone.', 'Day 3: Visit Garjia Temple.', 'Day 4: Stay overnight in the Dhikala forest rest house (if booked in advance).', 'Day 5: Morning safari from Dhikala and depart.'], '7-day': ['Day 1: Arrive, afternoon safari in Jhirna/Dhela zone.', 'Day 2:

Morning safari in Bijrani zone.', 'Day 3: Visit Garjia Temple.', 'Day 4: Stay overnight in the Dhikala forest rest house (if booked in advance).', 'Day 5: Morning safari from Dhikala.', 'Day 6: Visit the Corbett Waterfall.', 'Day 7: Explore the Corbett Museum at Kaladhungi.']}},
'Spiti Valley': {'budget': 50, 'climate': 15, 'interest': 90, 'weather_city': 'Kaza', 'itinerary': {'3-day': ['Day 1: Arrive in Kaza, acclimatize.', 'Day 2: Visit Key Monastery and Kibber village.', "Day 3: Visit the world's highest post office in Hikkim."], '5-day': ['Day 1: Arrive in Kaza, acclimatize.', 'Day 2: Visit Key Monastery and Kibber village.', "Day 3: Visit the world's highest post office in Hikkim.", 'Day 4: Visit Langza village (Buddha statue) and Komic village.', 'Day 5: Explore the Tabo Monastery.'], '7-day': ['Day 1: Arrive in Kaza, acclimatize.', 'Day 2: Visit Key Monastery and Kibber village.', "Day 3: Visit the world's highest post office in Hikkim.", 'Day 4: Visit Langza village (Buddha statue) and Komic village.', 'Day 5: Explore the Tabo Monastery.', 'Day 6: Visit Dhankar Monastery and Lake.', 'Day 7: Drive towards Chandratal Lake (seasonal).']}},
'Munnar': {'budget': 40, 'climate': 50, 'interest': 75, 'weather_city': 'Munnar', 'itinerary': {'3-day': ['Day 1: Arrive, visit a tea plantation.', 'Day 2: Visit Mattupetty Dam and Echo Point.', 'Day 3: Explore the Tea Museum.'], '5-day': ['Day 1: Arrive, visit a tea plantation.', 'Day 2: Visit Mattupetty Dam and Echo Point.', 'Day 3: Explore the Tea Museum.', 'Day 4: Visit Eravikulam National Park (for Nilgiri Tahr).', 'Day 5: Visit Top Station for views.'], '7-day': ['Day 1: Arrive, visit a tea plantation.', 'Day 2: Visit Mattupetty Dam and Echo Point.', 'Day 3: Explore the Tea Museum.', 'Day 4: Visit Eravikulam National Park (for Nilgiri Tahr).', 'Day 5: Visit Top Station for views.', "Day 6: Day trip to Kolukkumalai Tea Estate (world's highest).", 'Day 7: Relax and enjoy the scenery.']}},
'Yercaud': {'budget': 35, 'climate': 60, 'interest': 60, 'weather_city': 'Yercaud', 'itinerary': {'3-day': ['Day 1: Arrive, boat on Yercaud Lake (Emerald Lake).', "Day 2: Visit Lady's Seat viewpoint.", 'Day 3: Explore the Killiyur Falls.'], '5-day': ['Day 1: Arrive, boat on Yercaud Lake (Emerald Lake).', "Day 2: Visit Lady's Seat viewpoint.", 'Day 3: Explore the Killiyur Falls.', 'Day 4: Visit the Pagoda Point.', 'Day 5: Explore the Botanical Garden.'], '7-day': ['Day 1: Arrive, boat on Yercaud Lake (Emerald Lake).', "Day 2: Visit Lady's Seat viewpoint.", 'Day 3: Explore the Killiyur Falls.', 'Day 4: Visit the Pagoda Point.', 'Day 5: Explore the Botanical Garden.', 'Day 6: Visit the Servarayan Temple.', 'Day 7: Relax at a coffee estate.']}},
'Lavasa': {'budget': 45, 'climate': 65, 'interest': 50, 'weather_city': 'Lavasa', 'itinerary': {'3-day': ['Day 1: Arrive, walk along the lakeside promenade.', 'Day 2: Try some water sports on the lake.', 'Day 3: Relax at a cafe with a view.'], '5-day': ['Day 1: Arrive, walk along the lakeside promenade.', 'Day 2: Try some water sports on the lake.', 'Day 3: Relax at a cafe with a view.', 'Day 4: Go for a nature trail walk.', 'Day 5: Enjoy the planned city architecture.'], '7-day': ['Day 1: Arrive, walk along the lakeside promenade.', 'Day 2: Try some water sports on the lake.', 'Day 3: Relax at a cafe with a view.', 'Day 4: Go for a nature trail walk.', 'Day 5: Enjoy the planned city architecture.', 'Day 6: Relax and unwind.', 'Day 7: Enjoy the peaceful environment before departure.']}},
'Havelock Island': {'budget': 70, 'climate': 85, 'interest': 85, 'weather_city': 'Havelock', 'itinerary': {'3-day': ['Day 1: Arrive, visit Radhanagar Beach for sunset.', 'Day 2: Snorkeling or Scuba at Elephant Beach.', 'Day 3: Relax at Kalapathar Beach.'], '5-day': ['Day 1: Arrive, visit Radhanagar Beach for sunset.', 'Day 2: Snorkeling or Scuba at Elephant Beach.', 'Day 3: Relax at Kalapathar Beach.', 'Day 4: Try kayaking through the mangroves.', 'Day 5: Explore the local market and cafes.'], '7-day': ['Day

```
1: Arrive, visit Radhanagar Beach for sunset.', 'Day 2: Snorkeling or Scuba at Elephant Beach.', 'Day 3: Relax at Kalapathar Beach.', 'Day 4: Try kayaking through the mangroves.', 'Day 5: Explore the local market and cafes.', 'Day 6: Take a certified scuba diving course (PADI).', 'Day 7: Enjoy a final beach day before departing.']}},
    'Rameswaram': {'budget': 40, 'climate': 90, 'interest': 70, 'weather_city': 'Rameswaram', 'itinerary': {'3-day': ['Day 1: Visit the Ramanathaswamy Temple.', 'Day 2: Visit Dhanushkodi (Ghost Town).', 'Day 3: See the Pamban Bridge.'], '5-day': ['Day 1: Visit the Ramanathaswamy Temple.', 'Day 2: Visit Dhanushkodi (Ghost Town).', 'Day 3: See the Pamban Bridge.', 'Day 4: Visit the APJ Abdul Kalam Memorial.', 'Day 5: Relax at Agnitheertham beach.'], '7-day': ['Day 1: Visit the Ramanathaswamy Temple.', 'Day 2: Visit Dhanushkodi (Ghost Town).', 'Day 3: See the Pamban Bridge.', 'Day 4: Visit the APJ Abdul Kalam Memorial.', 'Day 5: Relax at Agnitheertham beach.', 'Day 6: Explore the Five-faced Hanuman Temple.', 'Day 7: Revisit the main temple for a different ceremony.']}},
    'Tirupati': {'budget': 30, 'climate': 85, 'interest': 65, 'weather_city': 'Tirupati', 'itinerary': {'3-day': ['Day 1: Arrive, prepare for temple visit.', 'Day 2: Darshan at Tirumala (Venkateswara Temple).', 'Day 3: Visit other local temples like Padmavathi Ammavari Temple.'], '5-day': ['Day 1: Arrive, prepare for temple visit.', 'Day 2: Darshan at Tirumala (Venkateswara Temple).', 'Day 3: Visit other local temples like Padmavathi Ammavari Temple.', 'Day 4: Visit the Sri Vari Museum.', 'Day 5: Explore the Talakona Waterfalls.'], '7-day': ['Day 1: Arrive, prepare for temple visit.', 'Day 2: Darshan at Tirumala (Venkateswara Temple).', 'Day 3: Visit other local temples like Padmavathi Ammavari Temple.', 'Day 4: Visit the Sri Vari Museum.', 'Day 5: Explore the Talakona Waterfalls.', 'Day 6: Visit the Deer Park.', 'Day 7: Relax before departure.']}},
    'Varanasi': {'budget': 35, 'climate': 80, 'interest': 70, 'weather_city': 'Varanasi', 'itinerary': {'3-day': ['Day 1: Arrive, evening Ganga Aarti at Dashashwamedh Ghat.', 'Day 2: Morning boat ride on the Ganges, explore the ghats.', 'Day 3: Visit Sarnath, where Buddha gave his first sermon.'], '5-day': ['Day 1: Arrive, evening Ganga Aarti at Dashashwamedh Ghat.', 'Day 2: Morning boat ride on the Ganges, explore the ghats.', 'Day 3: Visit Sarnath, where Buddha gave his first sermon.', 'Day 4: Explore the narrow lanes and temples of the old city.', 'Day 5: Visit the Banaras Hindu University campus.'], '7-day': ['Day 1: Arrive, evening Ganga Aarti at Dashashwamedh Ghat.', 'Day 2: Morning boat ride on the Ganges, explore the ghats.', 'Day 3: Visit Sarnath, where Buddha gave his first sermon.', 'Day 4: Explore the narrow lanes and temples of the old city.', 'Day 5: Visit the Banaras Hindu University campus.', 'Day 6: Attend a classical music or yoga session.', 'Day 7: Food tour to try local delicacies like kachori and lassi.']}}
}
destinations = base_destinations.copy()
API_KEY = "5d3b7fa8093c6037460a9f938d4d2190"

# --- Step 4: All Helper Functions ---
def map_budget_to_scale(budget_inr):
    min_budget, max_budget = 2000, 200000
    budget_inr = max(min_budget, min(budget_inr, max_budget))
    return ((budget_inr - min_budget) / (max_budget - min_budget)) * 100

def map_climate_to_scale(climate_celsius):
    min_climate, max_climate = 2, 60
```

```
    climate_celsius = max(min_climate, min(climate_celsius, max_climate))
    return ((climate_celsius - min_climate) / (max_climate - min_climate)) * 100

def simulate_backpropagation(disliked_place, user_prefs, state):
    nudge = 5
    current_destinations = state["destinations"]
    for attr, user_value in user_prefs.items():
        if attr in current_destinations[disliked_place]:
            place_value = current_destinations[disliked_place][attr]
            if user_value > 50 and place_value > 50:
                current_destinations[disliked_place][attr] = max(0, place_value - nudge)
            elif user_value < 50 and place_value < 50:
                current_destinations[disliked_place][attr] = min(100, place_value + nudge)
    state["destinations"] = current_destinations

def fuzzy_recommendation(budget, climate, interest, excluded_places=[],
current_destinations=None):
    if current_destinations is None: current_destinations = destinations
    scores = {}
    x_range = np.arange(0, 101, 1)
    for place, attrs in current_destinations.items():
        if place in excluded_places: continue
        b_score = fuzz.interp_membership(x_range, fuzz.trimf(x_range, [attrs['budget'] -
20, attrs['budget'], attrs['budget'] + 20]), budget)
        c_score = fuzz.interp_membership(x_range, fuzz.trimf(x_range, [attrs['climate'] -
20, attrs['climate'], attrs['climate'] + 20]), climate)
        i_score = fuzz.interp_membership(x_range, fuzz.trimf(x_range, [attrs['interest'] -
20, attrs['interest'], attrs['interest'] + 20]), interest)
        scores[place] = b_score + c_score + i_score
    if not scores: return None
    return sorted(scores.items(), key=lambda x: x[1], reverse=True)[0][0]

def get_weather(place):
    city_for_api = base_destinations.get(place, {}).get('weather_city', place)
    url =
f"http://api.openweathermap.org/data/2.5/weather?q={city_for_api}&appid={API_K
EY}&units=metric"
    try:
        res = requests.get(url, timeout=5).json()
        if res.get("cod") != 200:
            return f"❌ Weather info unavailable for {place} (API error:
{res.get('message', 'Unknown')})."
        return f"🌧️ Currently, it's **{res['main']['temp']}°C** with
{res['weather'][0]['description']} in the {city_for_api} area."
    except requests.exceptions.RequestException: return "❌ Network error."

def get_forecast(place):
    city_for_api = base_destinations.get(place, {}).get('weather_city', place)
```

```
    url =
f"http://api.openweathermap.org/data/2.5/forecast?q={city_for_api}&appid={API_K
EY}&units=metric"
    try:
        res = requests.get(url, timeout=5).json()
        if res.get("cod") != "200": return f"❌ Forecast info unavailable for {place}."
        seen_days, forecast_lines = set(), []
        for item in res['list']:
            dt = datetime.datetime.fromtimestamp(item['dt'])
            date_str = dt.strftime("%A, %b %d")
            if date_str not in seen_days:
                forecast_lines.append(f"- **{date_str}:** {item['main']['temp']}°C,
{item['weather'][0]['description']}")
                seen_days.add(date_str)
            if len(seen_days) >= 5: break
        return f"🗓️ Here is the 5-day forecast for the **{city_for_api}** area:\n" +
"\n".join(forecast_lines)
    except requests.exceptions.RequestException: return "❌ Network error."

def get_stay_suggestions(place):
    query = quote_plus(f"hotels in {place}")
    return f"""🏨 Here are some links to find great stays in **{place}**:
- **Google Maps:** [Search on Google
Maps](https://www.google.com/maps/search/?api=1&query={query})
- **Booking.com:** [Search on
Booking.com](https://www.booking.com/searchresults.html?ss={quote_plus(place)})
- **Airbnb:** [Search on
Airbnb](https://www.airbnb.com/s/{quote_plus(place)}/homes)"""

def display_itinerary(place, days):
    plan_key = f'{days}-day'
    if 'itinerary' in base_destinations[place] and plan_key in
base_destinations[place]['itinerary']:
        plan = base_destinations[place]['itinerary'][plan_key]
        response = f"🗓️ Here's a custom **{days}-Day Itinerary** for **{place}**:\n"
+ "\n".join([f"- **{day_plan}**" for day_plan in plan])
        return response
    return f"Sorry, I don't have a detailed {days}-day plan for {place} right now."

# --- Step 5: The Main Chatbot Logic for Gradio ---
def init_state():
    return {"step": "start", "budget": None, "climate": None, "interest": None,
"chosen_place": None, "disliked_places": [], "destinations": base_destinations.copy()}

def chat_response(message, history, state):
    if state is None: state = init_state()
    step, bot_message = state["step"], ""
```

```
    options_text = "\n\nWhat's next? You can ask for `forecast`, `stays`, or an itinerary (`3-day`, `5-day`, `7-day`). You can also say `new` to start over or `exit` to end."

    if message.strip().lower() in ["new", "start over", "reset"]:
        state = init_state()
        bot_message = "Ok, let's start over! 🗺️ \n\nWhat is your approximate **budget** in INR (e.g., 50000)?"
        state["step"] = "get_budget"
        return bot_message, state

    if step == "start":
        bot_message = "👋 Hi! I'm your Travel Assistant Chatbot.\n\nTo get started, what is your approximate **budget** in INR (e.g., 50000)?"
        state["step"] = "get_budget"
    elif step == "get_budget":
        try:
            val = int(message)
            state["budget"] = map_budget_to_scale(val)
            bot_message = f"Got it. A budget around ₹{val}.\n\nNow, what's your preferred **climate** in Celsius (e.g., 25)?"
            state["step"] = "get_climate"
        except ValueError: bot_message = "That doesn't look like a number. Please enter your budget in INR (e.g., 50000)."
    elif step == "get_climate":
        try:
            val = int(message)
            state["climate"] = map_climate_to_scale(val)
            bot_message = f"Perfect. A climate of {val}°C.\n\nLastly, on a scale of 0-100, how high is your interest in **adventure or heritage**?"
            state["step"] = "get_interest"
        except ValueError: bot_message = "That doesn't look like a number. Please enter your climate preference in Celsius (e.g., 25)."
    elif step == "get_interest":
        try:
            val = int(message)
            if 0 <= val <= 100:
                state["interest"] = val
                place = fuzzy_recommendation(state["budget"], state["climate"], state["interest"], state["disliked_places"], state["destinations"])
                if place:
                    state["chosen_place"] = place
                    bot_message = f"Based on your preferences, my recommendation is **{place}**!\n\n{get_weather(place)}\n\nDo you **like** this suggestion, or should I find **another** one?"
                    state["step"] = "feedback"
                else:
                    bot_message = "I couldn't find any more places. Would you like to **start over**?"
                    state["step"] = "start"
            else: bot_message = "Please enter a number between 0 and 100."
```

```
        except ValueError: bot_message = "That doesn't look like a number. Please enter
a value between 0 and 100."
    elif step == "feedback":
        feedback = message.lower()
        if "like" in feedback or "yes" in feedback:
            bot_message = f"Great! What would you like to know about
**{state['chosen_place']}**?" + options_text
            state["step"] = "explore"
        elif "another" in feedback or "no" in feedback:
            state["disliked_places"].append(state["chosen_place"])
            user_prefs = {'budget': state["budget"], 'climate': state["climate"], 'interest':
state["interest"]}
            simulate_backpropagation(state["chosen_place"], user_prefs, state)
            place = fuzzy_recommendation(state["budget"], state["climate"],
state["interest"], state["disliked_places"], state["destinations"])
            if place:
                state["chosen_place"] = place
                bot_message = f"No problem. Let's try:
**{place}**!\n\n{get_weather(place)}\n\nDo you **like** this suggestion, or want
**another** one?"
                state["step"] = "feedback"
            else:
                bot_message = "I've run out of recommendations. Would you like to **start
over**?"
                state["step"] = "start"
        else: bot_message = "Sorry, I didn't understand. Please tell me if you **like**
the suggestion or want **another** one."

    elif step == "explore":
        query, place = message.lower(), state["chosen_place"]

        if "forecast" in query:
            bot_message = get_forecast(place) + options_text
        elif "stay" in query:
            bot_message = get_stay_suggestions(place) + options_text
        elif "3" in query:
            bot_message = display_itinerary(place, 3) + options_text
        elif "5" in query:
            bot_message = display_itinerary(place, 5) + options_text
        elif "7" in query:
            bot_message = display_itinerary(place, 7) + options_text
        elif "exit" in query:
            bot_message = "Wishing you a safe and wonderful journey! Feel free to talk to
me again anytime. Goodbye! 👋"
            state = init_state()
        else:
            bot_message = f"Sorry, I didn't catch that. For **{place}**, I can help with
`forecast`, `stays`, or itineraries." + options_text

    time.sleep(1)
```

```
    return bot_message, state

# --- Step 6: Launch the Gradio Chat Interface ---
with gr.Blocks(theme=gr.themes.Soft(), title="Travel Assistant") as demo:
    gr.Markdown("## Travel Assistant ChatBot")
    gr.Markdown("Your personal guide to discovering the perfect travel destination in
India. Say 'new' or 'reset' at any time to start over.")
    chatbot_state = gr.State()
    gr.ChatInterface(fn=chat_response, additional_inputs=[chatbot_state],
additional_outputs=[chatbot_state], textbox=gr.Textbox(placeholder="Your
personalized journey across India starts here! ✨ Let's plan your
trip.",autofocus=True), title="Travel Chatbot")

demo.launch(debug=True, share=True)
```